

Open Risk Map

River Deep Mountain AI

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Abstract

The **Open Risk Map** is developed by **River Deep Mountain AI** (RDMAI)¹, a collaborative innovation project funded by the **Ofwat Innovation Fund**. RDMAI aims to develop open-source AI/ML models to inform effective waterbody pollution management. The Open Risk Map uses openly-available data to map water quality ‘risk’ across catchments in England. Two pollutants are represented: orthophosphate as P and ammoniacal nitrogen as N. The maps can be extended to represent additional pollutants.

Open water quality data are available from sources such as **Environment Agency** (EA) and can be used to inform understanding of the risk of river water pollution. Although data is available, there is no easy method to visualize how water quality parameters vary in a catchment. The Open Risk Map aims to provide a clear method of mapping these data together to show the spatial distribution of water quality risk.

The tool collates and combines data from multiple open data sources to produce a ‘risk map’ comprising multiple pollution source layers. The map apportions pollution to specific sources in a catchment, and a pollution risk at each point in the catchment river network is estimated. The tool provides an interactive, visual demonstration of how pollution risk varies from throughout a catchment river network.

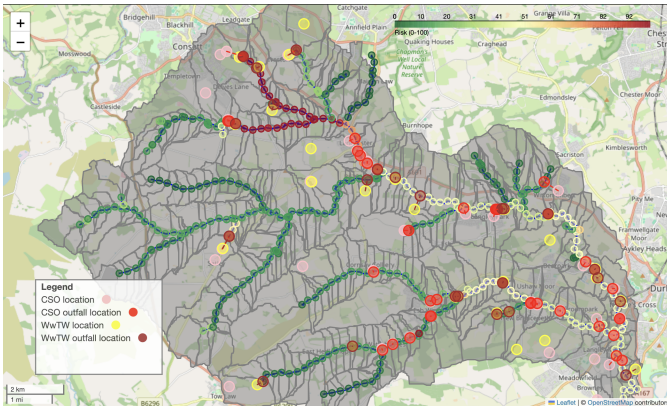


Figure 1: Risk map for the **River Browney** catchment (Durham, UK)

Fig. 1 shows the risk map of the River Browney catchment

¹The project consists of 6 core partners: Northumbrian Water, Cognizant Ocean, Xylem Inc., Water Research Centre Limited, The Rivers Trust and ADAS. The project is further supported by 6 water companies across the United Kingdom and Ireland.

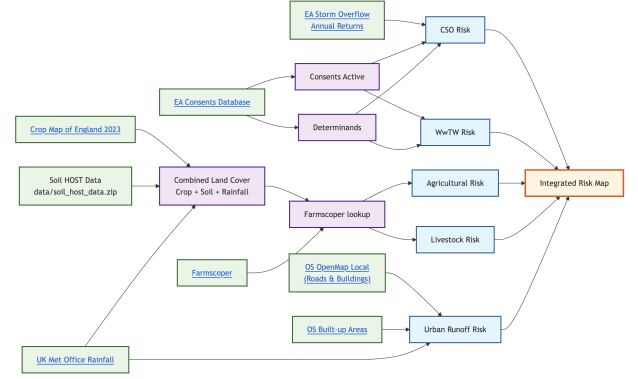


Figure 2: Data flow used to generate the risk map

(1819 *ha*, County Durham), visualising both orthophosphate (*P*) and ammoniacal nitrogen (*N*) risk, with a value above 100 defined as ‘poor’ according to the standard chosen for that pollutant. The various Wastewater Treatment Works (WwTWs) and Combined Sewer Overflows (CSOs) can also be seen on the catchment map.

1 DATA SOURCES

Data loading and transformation is a key part of the model. Multiple data sources are combined to form the individual risk layers (Fig. 2). A detailed description of the data extraction and transformation can be found in the project’s [github repository](#).

The river network geometry and topology is obtained from the **Open River Network** (ORN)^[1]. Data for the duration and location of CSOs are obtained from the **EA Storm Overflows - Annual Returns**. Consented discharges of WwTWs and CSOs are also obtained from the EA. Agricultural land use data are obtained from the **Crop Map of England 2023 (CROME)** dataset. Soil data mapping Hydrology of Soil Types (HOST) classes are taken, with permission, from Schneider et al.^[2]. Rainfall data are obtained from the **UK MET Office**. Topography data are obtained using the **elevatr** python library. The Ordnance Survey **local maps** and **built-up areas map** provide data for roads and buildings. Soil runoff load data are obtained from the **Farmscoper Tool** developed by **ADAS**.

2 MODEL ARCHITECTURE

The model represents the river network as a directed [NetworkX](#) graph, with the ORN providing flow direction. Nodes are placed at regular intervals along the river network according to user preference (e.g. every 100m).

2.1 RIVER FLOW RATE

Estimating pollutant dilution at different points in the river network requires an estimation of river flow (m^3/s). The risk map implements a deliberately simple flow estimation, with the anticipation of future integration of the risk map model with specialised flow estimation models, such as the [RDMAI Open Flow Model](#). The flow estimation uses data from the [National River Flow Archive](#) (NRFA) gauging sites. A linear correlation between ORN upstream accumulated river length and median flow is established. ($n = 1481, r^2 = 0.7$)

$$\dot{V} = 1.476 \times 10^{-5} \times x$$

Where $\dot{V}$ is the median river flow rate (m^3/s) throughout the year and x is the upstream accumulated river length (m) - the total river length upstream of the point.

2.2 MAPPING POINT SOURCES

Mapping point sources to the river network is not trivial as they often lie some distance from the river itself. Additionally, the ORN dataset represents the river network as lines, with no indication of river width, which is often considerable. Hence, point sources are mapped to the river network at the closest edge within a certain maximum distance threshold (700m) - chosen to account for the largest river widths (Fig. 3a).

After mapping, pollutant loading from each point source and its distance from the river network are used to calculate a value for risk at every point along the river network.

2.3 MAPPING SPATIALLY EXTENDED SOURCES

Representing extended / diffuse pollution sources in the risk map requires the delineation of catchment surface area contributing runoff to each node in the river network. This is achieved using the [whitebox-workflows](#)^[3] python library (for calculating sub-catchments) which is run using topographical data from the elevatr library.

Due to an artifact of the resolution of the topographical data, some delineated catchments can be very small. Catchments with an area less than $20,000m^2$ are combined with the downstream catchments, with pollutant load divided among them (Fig. 3b).

3 LOAD ESTIMATION

The risk map is designed to estimate the long-term, annual average spatial distribution of river pollution risk, partly due to the limitations of the datasets used. We therefore

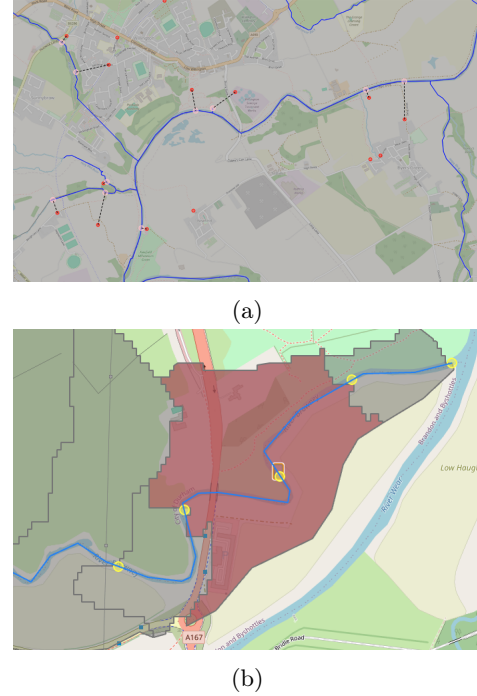


Figure 3: **a** - An illustration of mapping point sources to the river network. Different pollution sources (red points) are mapped to the river network (blue) and assigned assumed outfall locations (pink). **b** - A small sub-catchment (central yellow rectangle) is identified as below the area threshold and is combined with the larger sub-catchment (red).

estimate annual average pollutant loads from the different sources (WwTWs, CSOs, urban and agricultural runoff) used in the model.

3.1 WASTEWATER TREATMENT WORKS (WwTWs)

WwTWs are point sources. In England, the consented discharges database from the EA lists WwTW consents for effluent concentrations of different pollutants. In the risk map, these consents are used to estimate both the WwTW effluent concentration and flow rate. The *Dry Weather Flow* (DWF) is used as a substitute for the effluent flow, while the consents for *reactive orthophosphate* load or *ammoniacal nitrogen* load are used for the final effluent pollutant concentration.

Many WwTWs are not consented for effluent flow and/or final effluent concentration, in which case, assumptions are made to infill missing data:

- Using the EA’s [Water Quality Archive](#), the mean orthophosphate (*P*) concentration is estimated for storm sewage discharge samples at 5mg/l ($n = 354, 370$). For ammoniacal N, a value of 2.6mg/l ($n = 276, 362$) is used.
- The DWF is assumed to be $20\text{m}^3/\text{day}$. A manual check of sites without a DWF consent indicated the sites to generally be very small. The assumed DWF value is the upper discharge limit of the ‘[Standard Rules](#)’ permit by the EA.

In order to represent private WwTWs, a modification of the above assumptions was required:

- Domestic Property (Single): These are assumed to have a DWF of $0.8\text{m}^3/\text{day}$ ($800\text{l}/\text{day}$), derived as the median of their DWF consents (mean = 1.72, SD = 3.64, $n = 361$).
- Domestic Property (Multiple): These are assumed to have a DWF of $7\text{m}^3/\text{day}$ ($7000\text{l}/\text{day}$) derived similarly ($n = 133$).

3.2 COMBINED SEWER OVERFLOWS (CSOs)

While there are many types of overflows, we use the common terminology of ‘Combined Sewer Overflows’ (CSOs). The risk map uses the consents data to estimate the concentration and the spill volume of CSOs. *Weir Setting* is used as a substitute for spill volume. However, since CSOs are not generally consented for pollutant concentrations, they were assumed to be the same for all CSOs.

- Samples of CSO outfall in the EA Water Quality Archive indicated a mean Orthophosphate as P of 2.42mg/l ($n = 674, 243$). This value may seem low, but overflow discharges are generally diluted by surface runoff. With this in mind, a final value of 3mg/l was used. Similarly, the mean ammoniacal N concentration of 13mg/l ($n = 254$) was used.

- Since the Weir Setting varies considerably in unconsented CSOs, a CSO spill rate of $20\text{l}/\text{s}$ was assumed.

Spill durations also need to be considered when estimating yearly CSO pollutant loads. The total spill duration of each CSO provided by the EA Storm Overflows dataset was used to calculate the average load of the CSO over the year.

$$L_{avg} = \frac{t_s}{365 \times 24 \times 60 \times 60} L$$

Where L is the estimated load per second during a spill and $t_s[s]$ is the annual spill duration of that CSO.

3.3 AGRICULTURAL RUNOFF

The risk maps only consider spatially extended source pollution originating from agriculture. This consists of 2 components: crop agriculture and livestock. The Farmscoper tool is used to estimate the pollutant loads from both. The tool outputs lookup tables which provide a yearly load for the pollutant per unit area.

3.3.1 CROP AGRICULTURE

Estimating pollutant load from crop agriculture requires a *land use category*, obtained from the CROME dataset, a *rainfall category*, obtained from the MET Office, and a *soil category*, obtained from the soil HOST classes. This gives us the crop agriculture pollutant load for each land parcel.

3.3.2 LIVESTOCK

The numbers of different livestock animals are estimated across many catchments in the UK, as per the [2019 agricultural census data](#). These data are used to estimate livestock excreta per hectare in a land parcel. This information is combined with the *land use category*, *soil category* and *rainfall category* to give the pollutant load for each land parcel.

3.3.3 INTERCEPTION AND RE-INFILTRATION OF RUNOFF

Agricultural pollution enters the river network through 2 pathways: subsurface flow and surface runoff. Pollutant load in subsurface flow is assumed to experience no degradation, whereas load in surface runoff is assumed to degrade with distance, estimated as:

$$L = L_0 e^{-\frac{x}{100}}$$

Where L_0 is the load and $x(m)$ is the closest distance to the river segment in the relevant (to the polygon) sub-catchment.

3.4 URBAN RUNOFF

Urban runoff is treated as another type of spatially extended pollution source. Two types of urban runoff have been considered: runoff from roads and runoff from buildings. Run-off from urban sources tends to have high concentrations of heavy metals, such as zinc and copper, or polycyclic aromatic hydrocarbons (PAH), such as benzo[a]pyrene (among a host of other contaminants). In contrast, urban runoff is considered to contribute to relatively minor loads of orthophosphate P and ammoniacal N, although these contributions can be significant in some urban watercourses. Therefore, the risk maps do not consider traffic data. Instead, pollutant loads from urban runoff are estimated using rainfall and event mean concentrations. Here, total area of the delineated catchment is multiplied by the annual rainfall and the event mean concentration.

3.4.1 ROADS

As the Ordnance Survey dataset represent roads as lines with no indication of width, road widths were estimated using a lookup table. This table was constructed by taking samples of each *road classification*, and then measuring the widths of these roads in [Google Earth](#) using the path tool, with the table listing the average width for each road classification. Roads are then categorised as rural or urban using the built-up areas map. Rural roads which have been further categorised as trunk roads are highlighted when they are identified as being within 200m of the river network. For Active P, a value of 0.1mg/l is used as the event mean concentration for both urban and rural trunk roads. For Ammoniacal N, a value of 0.8mg/l is used for urban roads and 0.65mg/l for rural trunk roads. These values have been obtained using existing literature^[4,5,6,7] and unpublished data from [Prof. Werner](#) at Newcastle University.

3.4.2 BUILDINGS

Using the built-up areas map, buildings are separated into rural and urban. The pollutant load from rural buildings is not considered since it is assumed that runoff from most rural buildings will drain to soakaways, and have minimal impact. For orthophosphate P, an event mean concentration of 0.1mg/l is assumed according to Gikas and Tsihrantzis^[8]. Since a reasonable value for ammoniacal N could not be clearly identified from the literature, a value of 0.8mg/l was assumed - corresponding with the published mean event concentration for roads.

4 RISK CALCULATION

4.1 POLLUTANT DEGRADATION

After calculating pollution loads from the different sources discussed above, loadings are propagated downstream

along the river network. Orthophosphate P and ammoniacal N are non-conservative water quality variables, meaning they are known to degrade over time. This degradation is represented as:

$$L = L_0 e^{-kt} = L_0 e^{-\frac{kx}{u}}$$

Where L_0 is the load at a point and L is the load at the next point, $k(s^{-1})$ is the degradation coefficient, $x(m)$ is the distance between the 2 points, and $u(m/s)$ is the river flow velocity.

The risk map assumes a constant flow velocity of 0.77m/s, obtained as the mean flow velocity in 5 assessed case study catchments using the formulation in this report^[9]. A *Manning's roughness* of 0.03 is used along with the slope, width and flow rate from the NRFA gauging sites.

The degradation coefficients for orthophosphate P and ammoniacal N were obtained from WRc's in-house model [SIMPOL-ICM](#). Using a summer water temperature of 16°C, the degradation coefficients used for orthophosphate P and ammoniacal N are 0.04day⁻¹ and N is 0.42day⁻¹, respectively.

4.2 RISK VALUE

The average concentration $c_{avg}(mg/l)$ of a pollutant at a point in the river network can be obtained by dividing the combined load $L(g/s)$ from all the sources with the flow rate $\dot{V}(m^3/s)$. This concentration is then used to estimate the risk.

$$c_{avg} = \frac{\sum L}{\dot{V}}$$

- For orthophosphate P, the UKTAG Revised Standards^[10] are used, which define a river reach with a orthophosphate P concentration exceeding 0.87mg/l to be of a 'poor' water quality status.
- For ammoniacal N, the [Water Framework Directive](#) defines a river reach with a 99th percentile exceeding 6mg/l to be of a 'poor' water quality status.

The pollutant concentrations are then normalised to a risk value according to the above standards as percentages, with a risk value exceeding 100 indicating 'poor' river health.

$$R = \frac{c_{avg}}{c_{standard}} \times 100$$

Where R is the risk value and $c[mg/l]$ is concentration of the pollutant in the river.

5 VALIDATION

The risk map is not validated using the same metrics as a water quality model as it is not intended to perform as such. The intent is to perform a more qualitative assessment of the strengths and the shortcomings of the model and the contributing factors.

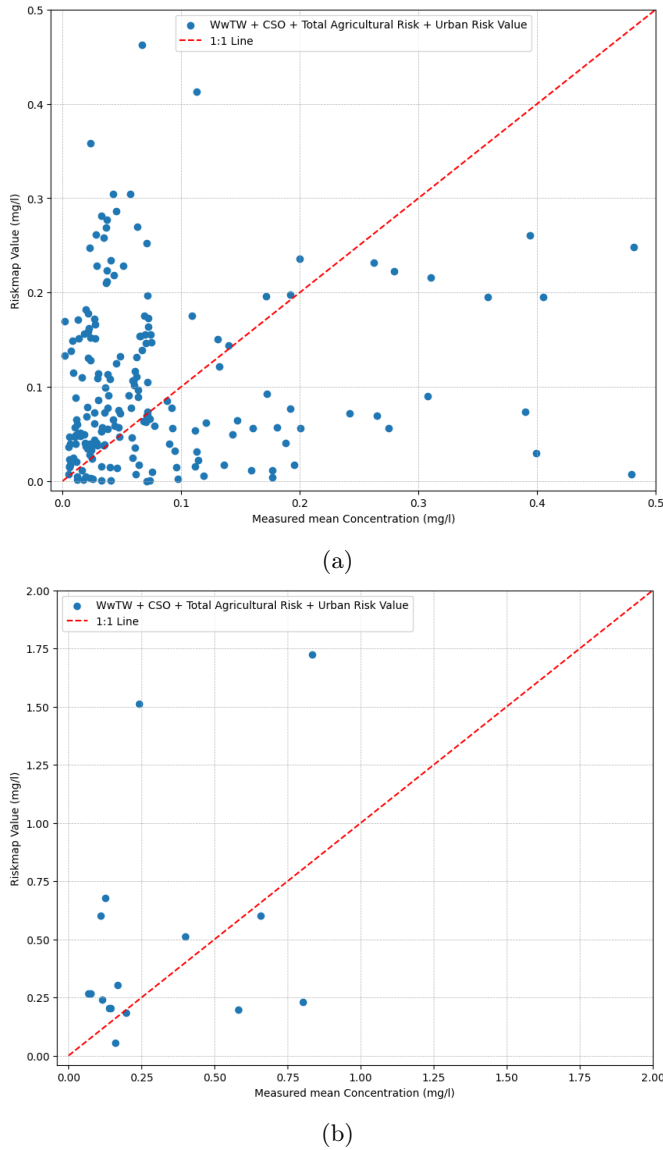


Figure 4: Combined predicted vs. observed mean values for orthophosphate P (a) and ammoniacal N (b). The red dotted line represents the 1 : 1 line.

5.1 ORTHOPHOSPHATE P

Measures of orthophosphate P from 2017 to 2024 in the EA Water Quality database were used in the validation. Only monitoring points with (> 20) measurements were considered.

The validation considered 11 catchments, chosen to be representative of the variety of catchments across England. As shown in Fig. 4a, the model generally over-predicted the concentration of orthophosphate P. This is expected, as sparse EA monitoring data is likely to miss spikes in pollution associated with ephemeral events, such as acute rainfall.

In the cases where risks are over predicted, risk from agriculture had the highest contribution, indicating some calibration of the agricultural pollutant load formulation

could be required.

5.1.1 CASE STUDY: AVON HAMPSHIRE

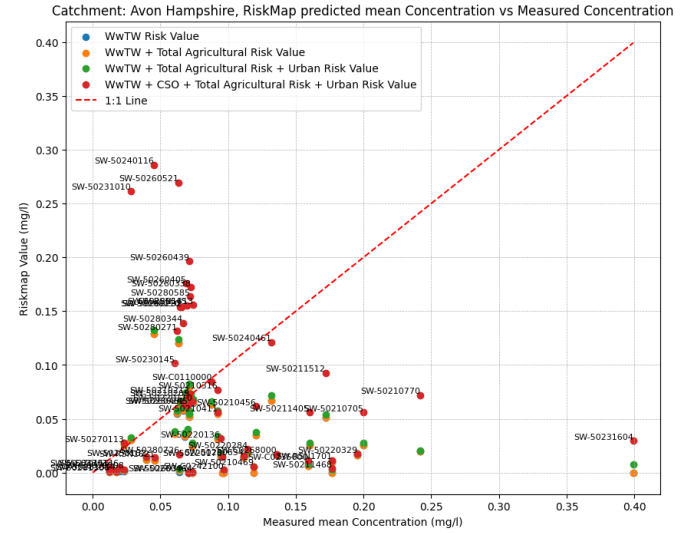


Figure 5: Predicted vs. actual mean concentrations of orthophosphate P for the Avon Hampshire catchment

In the Avon Hampshire catchment, most predicted orthophosphate P concentrations are $< 0.2 \text{ mg/l}$, representing low risk. Therefore, the impact of the over predictions is limited. However, in Fig 5, a cluster of unusually high values close to the Y-axis stand out. Two points from this cluster are chosen for further analysis by plotting the underlying orthophosphate P time series (Fig. 6a and Fig. 6b). The results show that CSOs and WwTWs represent the greatest contributions to risk at these points. Further analysis shows that CSOs and WwTWs contributing to pollutant loads at these points are being represented using assumed effluent flow rate and pollutant concentrations due to a lack of consent data. The inclusion of additional data into the risk map, such as water company owned data, can help avoid assumptions that lead to such over-predictions of water quality risk.

5.1.2 CASE STUDY: EDEN LOWER

Risk model predictions of orthophosphate P for the Eden Lower catchment were generally in good agreement with the EA measurements. However, the few points for which the maps over-estimate orthophosphate P have been further analysed in Fig. 8c and Fig. 8d. The model over prediction can be attributed to high pollutant loads from crop agriculture and livestock. These results indicate a possible need to re-visit the risk map formulation for estimating agricultural orthophosphate P loads.

5.2 AMMONIACAL N

Validation of the risk map estimations of ammoniacal N used data from the EA Hydrology Data Explorer. These

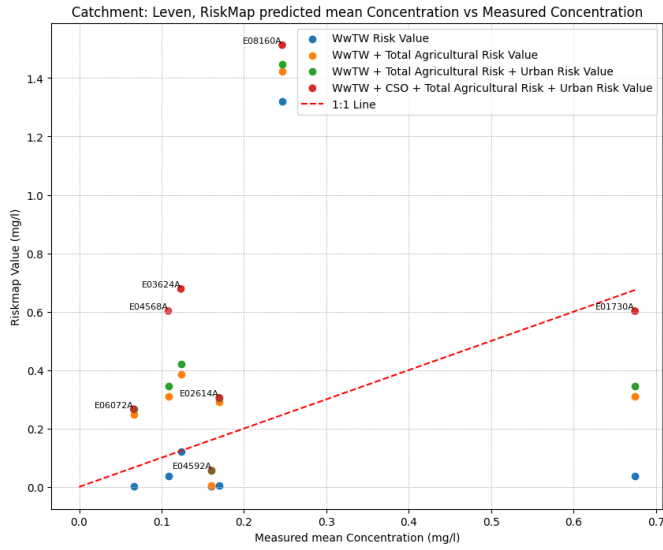


Figure 9: Risk map estimated vs EA measured mean concentrations of ammoniacal N for the Leven catchment

in Fig. 10b), one of the largest lakes in England. The risk maps do not account for standing water bodies. Therefore, this lake might invalidate many assumptions on river flow speed and pollutant degradation made in the risk map for this particular location. The other points in Fig. 10c and Fig. 10d show general good agreement between model estimated and EA measured ammoniacal N.

5.2.2 CASE STUDY: OUSE BURN

Measured EA ammoniacal N data is available for validation in the Ouse Burn (Newcastle) for June - October 2025. While this is a short time period, the validation can still provide some information on model performance. As shown in Fig. 11a, the risk map under-predicted ammoniacal N for this point.

Fig. 11b and 11c show that while the risk map captures the measured baseline ammoniacal N concentration, spikes in the measured concentration are not represented. However, it should be noted that the risk map produces a long-term annual average concentration, whereas the few months of sensor data available for this catchment will be influenced by acute events that result in short-term elevated pollution levels.

5.3 FLOW RATE

The risk map model implements a deliberately simple flow estimation in anticipation of later integration with a more sophisticated flow data source. The effect of this rudimentary flow estimation on overall model performance was assessed. Here, flow gauging stations near EA monitoring points containing orthophosphate P measures were used. The flow data from these gauging stations were used in the risk model instead of estimating flow, and the estimated orthophosphate P was then compared to EA mea-

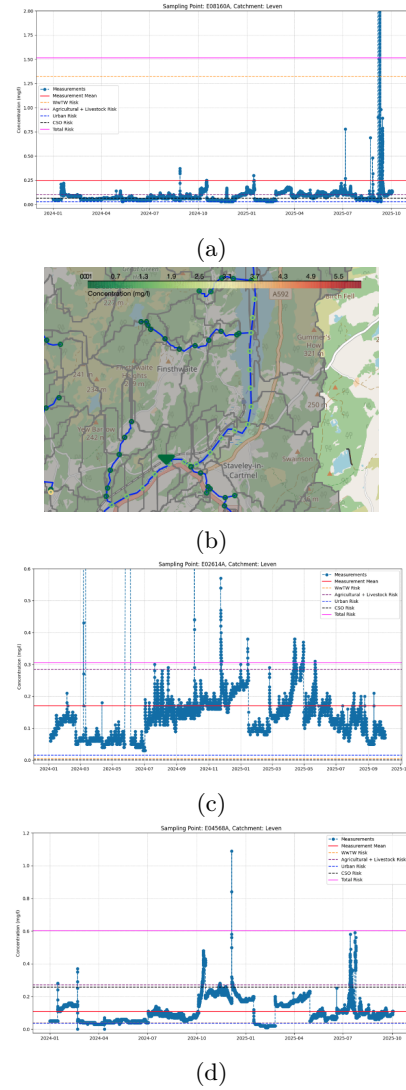


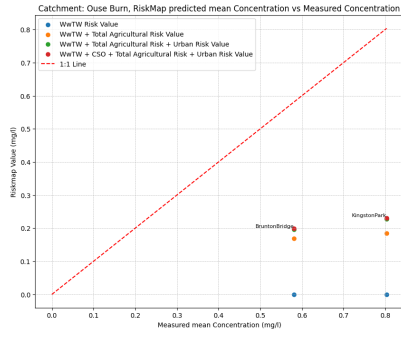
Figure 10: Time series for a, c and d. The location of a is shown in b to be downstream of Windemere.

surements.

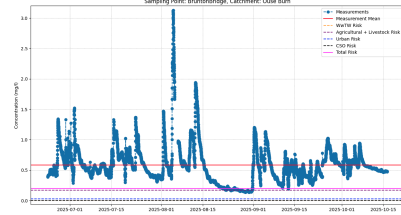
In Fig. 12, it can be seen that correcting for flow for these measurements does have a significant positive impact on the r^2 score of the prediction, going from $r^2 = -0.45$ to $r^2 = 0.55$ after correction.

6 BENCHMARKING

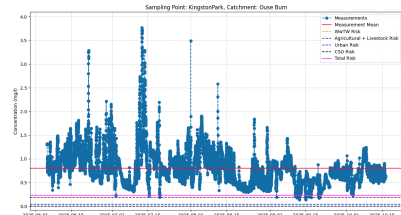
The risk map was compared to a similar, industry-standard tool to assess its relative performance, ease of use, accessibility, cost, etc. Risk map estimates of orthophosphate P for 12 catchments across England were compared to those of a UK Water Industry Research (UK-WIR) Source Apportionment Geographical Information Systems (SAGIS) dataset. The results showed that both models tended to over-estimate orthophosphate P compared to EA measurements for some catchments, with a similar overall performance. The SAGIS model generally generated higher orthophosphate loads from WwTWs and



(a)



(b)



(c)

Figure 11: Results for the Ouse Burn Catchment. (a) shows the combined results. (b) and (c) show the time series plots of the measured data.

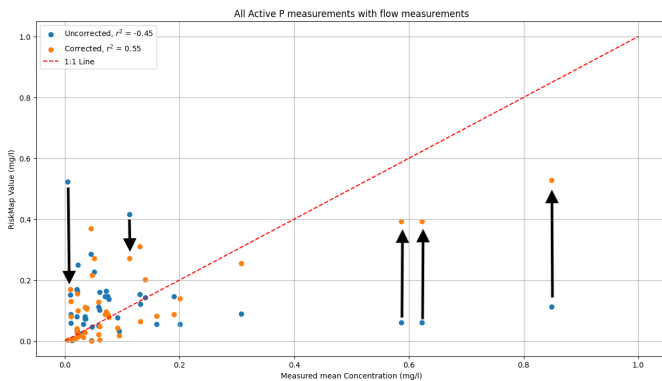


Figure 12: Plot of the predicted vs. measured mean concentrations of P with and without flow correction. The arrows indicate the movement of the values closer to the 1:1 line.

lower orthophosphate loads from CSOs than the risk map. This benchmarking process could not assess which model was more ‘correct’. The risk map offers definite advantages in ease of use, visualisation and accessibility, and the incorporation of improved flow data could significantly improve its performance.

7 CONCLUSIONS

Overall, the risk map has been successful in its goal of providing a map of catchment water pollution source apportionment and risk. A solid methodology has been developed which can be built upon by incorporating additional pollutants, pollution sources, or improved data sources or approaches. All data used is open, allowing easy augmentation with bespoke data, if available. As it is open source, the model can be readily adapted for specific use cases. These use cases can include:

- To allow catchment managers to identify likely contributors to a pollution event for targeted sampling.
- To help citizen science samplers or catchment managers design optimal monitoring campaigns.
- To assess how a new WwTW or CSO could affect water quality risk in a catchment.

The validation and benchmarking have also yielded insight into several areas where the model could be improved. Examples of these are:

- A better approximation of flow (such as the RDMAI’s [Open Flow Model](#)).
- Improved river flow speed approximation could provide a better indication of risk degradation.
- Improved assumptions for infilling missing information for WwTWs and CSOs.
- More accurate datasets could be used for road widths / pollution contributions to calculate urban runoff.
- The model could be made to learn from existing data using something like a Gaussian process, which could provide significantly more insight and the ability for continuous learning.

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The model described in this document is the product of an ongoing research and development project. It represents an early iteration and its outputs are preliminary. The model is not intended for use in production environments, regulatory reporting, environmental monitoring, or any decision-making in relation to public health, public policy, or commercial matters without independent validation and the acceptance of full liability by the user.

All use of the model and its associated software is subject to the full terms set out in the RDMAI repository, available at <https://github.com/Cognizant-RDMAI>.

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